

DAY 05

Ethernet LAN Switching

Part 1



38 original flashcards preserved

Visual focus: Ethernet frame fields • MAC addresses • EtherType • switching logic

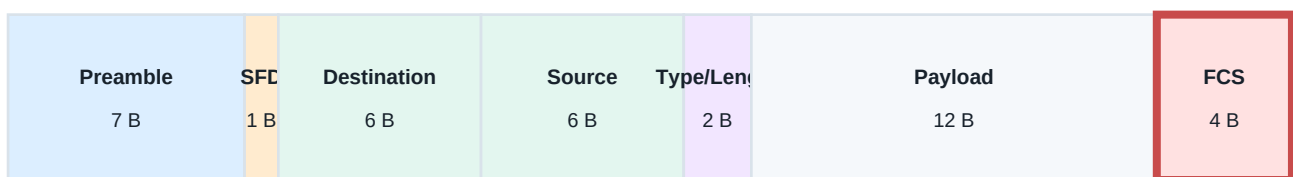
Every original question and answer is included, with added memory cues and diagrams.

Study method: cover the answer box, recall, then check the visual.

QUESTION

What is the only field in an Ethernet trailer?

VISUAL



Simplified Ethernet frame layout (payload width compressed for study)

ANSWER

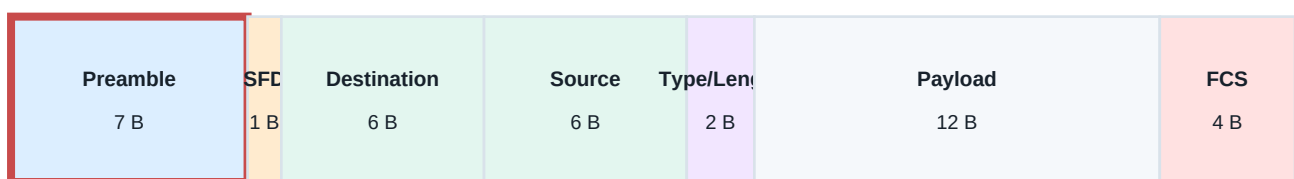
FCS (Frame Check Sequence)

Memory cue: The trailer is simple: it carries the FCS for error detection.

QUESTION

What are the 5 fields of an Ethernet header?

VISUAL



Simplified Ethernet frame layout (payload width compressed for study)

ANSWER

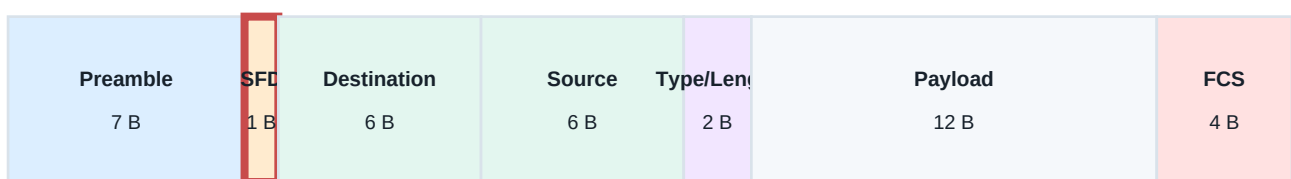
Preamble, SFD, Destination, Source, Type/Length

Memory cue: Header = Preamble + SFD + Destination + Source + Type/Length.

QUESTION

What does 'SFD' in an Ethernet header stand for?

VISUAL



Simplified Ethernet frame layout (payload width compressed for study)

ANSWER

Start Frame Delimiter

Memory cue: Connect this fact to the Ethernet frame and the switch MAC address table.

QUESTION

What does 'FCS' in an Ethernet frame stand for?

VISUAL

ETHERNET

Frame fields • MAC addressing • switching

ANSWER

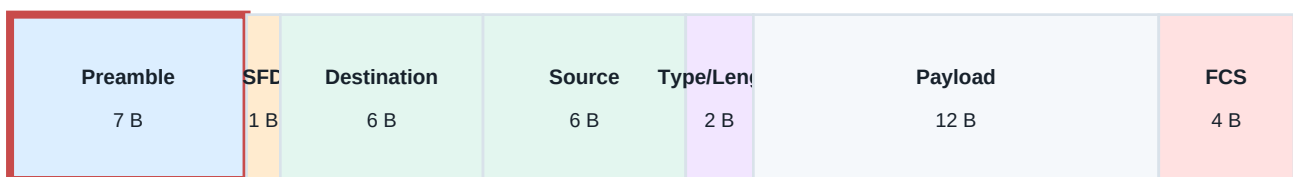
Frame Check Sequence

Memory cue: Connect this fact to the Ethernet frame and the switch MAC address table.

QUESTION

Ethernet header 'Preamble' = [...] byte(s)

VISUAL



Simplified Ethernet frame layout (payload width compressed for study)

ANSWER

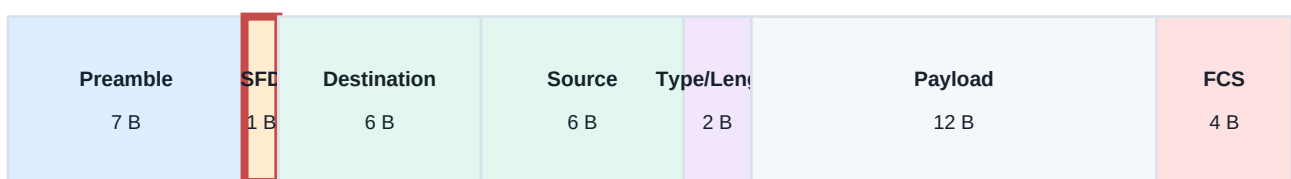
7

Memory cue: Preamble is 7 bytes; SFD adds 1 more byte before the destination MAC.

QUESTION

Ethernet header 'SFD' = [...] byte(s)

VISUAL



Simplified Ethernet frame layout (payload width compressed for study)

ANSWER

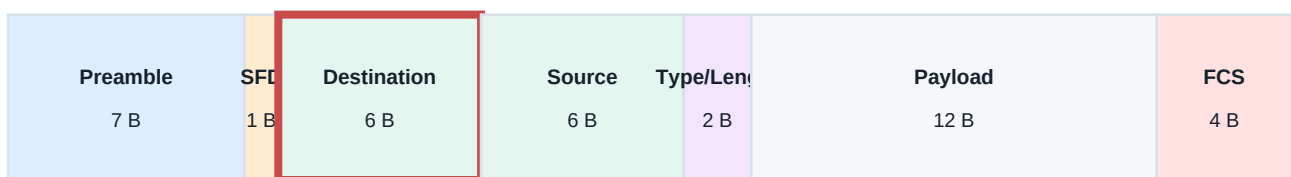
1

Memory cue: SFD is exactly 1 byte and marks the end of the preamble.

QUESTION

Ethernet header 'Destination' = [...] byte(s)

VISUAL



Simplified Ethernet frame layout (payload width compressed for study)

ANSWER

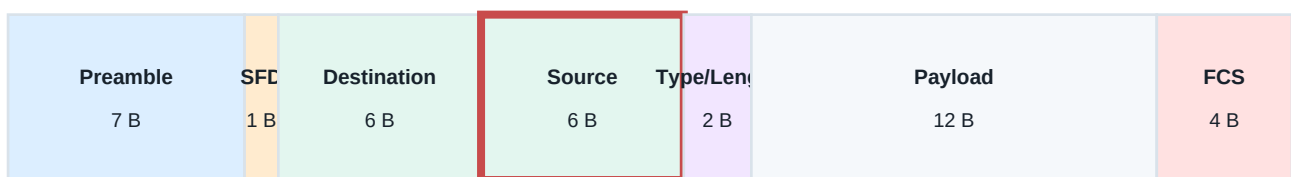
6

Memory cue: Destination MAC = 6 bytes = 48 bits.

QUESTION

Ethernet header 'Source' = [...] byte(s)

VISUAL



Simplified Ethernet frame layout (payload width compressed for study)

ANSWER

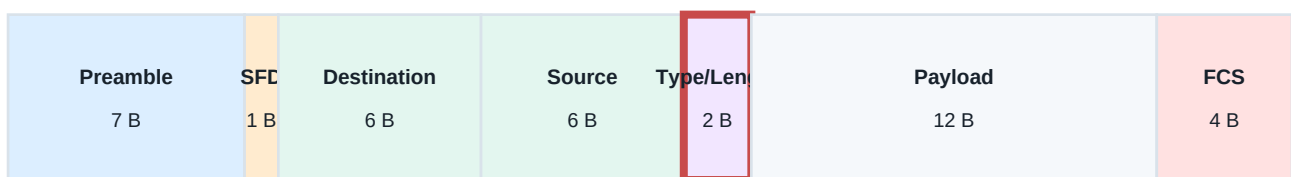
6

Memory cue: Source MAC = 6 bytes; switches learn from this field.

QUESTION

Ethernet header 'Type/Length' = [...] byte(s)

VISUAL



Simplified Ethernet frame layout (payload width compressed for study)

ANSWER

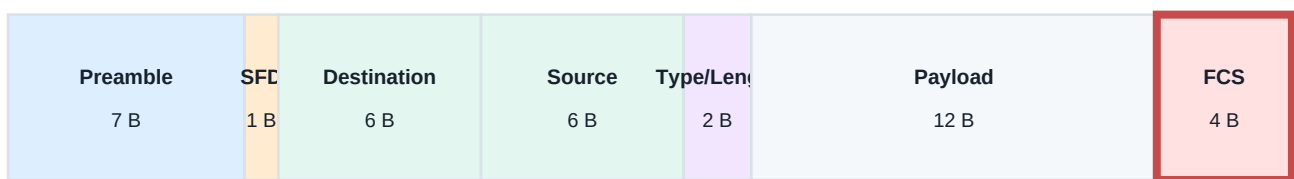
2

Memory cue: Type/Length is 2 bytes and tells either payload length or upper-layer protocol.

QUESTION

Ethernet trailer 'FCS' = [...] byte(s)

VISUAL



Simplified Ethernet frame layout (payload width compressed for study)

ANSWER

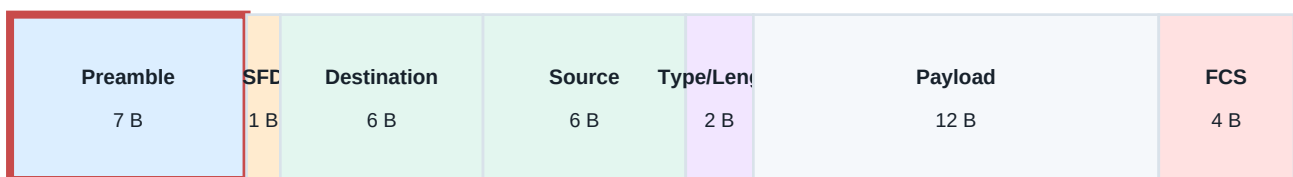
4

Memory cue: FCS is 4 bytes and uses CRC to detect corruption.

QUESTION

Which field of an Ethernet header is used for synchronization?

VISUAL



Simplified Ethernet frame layout (payload width compressed for study)

ANSWER

Preamble

Memory cue: Think: preamble helps receiver timing before the real frame fields begin.

QUESTION

What is the bit pattern of one byte in the Ethernet preamble?

VISUAL



One byte shown bit-by-bit

ANSWER

10101010

Memory cue: Preamble repeats 10; SFD ends with 11.

QUESTION

What is the bit pattern of the Ethernet SFD?

VISUAL



One byte shown bit-by-bit

ANSWER

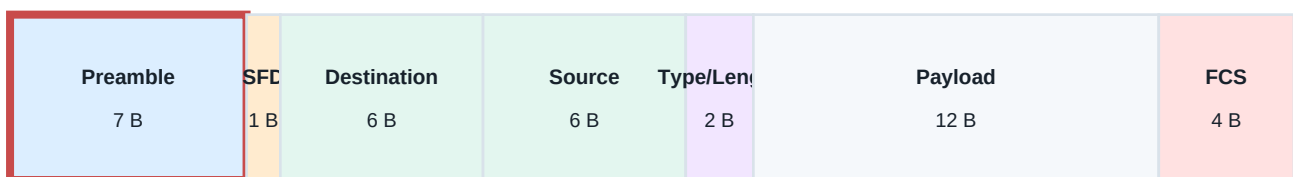
10101011

Memory cue: Preamble repeats 10; SFD ends with 11.

QUESTION

Which field of an Ethernet header marks the end of the preamble?

VISUAL



Simplified Ethernet frame layout (payload width compressed for study)

ANSWER

SFD

Memory cue: SFD = the boundary between preamble and destination MAC.

QUESTION

What does MAC stand for?

VISUAL



6 bytes = 48 bits = 12 hexadecimal characters

ANSWER

Media Access Control

Memory cue: MAC is the Layer-2 addressing concept used by Ethernet switches.

QUESTION

How long is a MAC address? = [...] bytes

VISUAL



6 bytes = 48 bits = 12 hexadecimal characters

ANSWER

6

Memory cue: 6 bytes $\times$ 8 bits = 48 bits.

QUESTION

How long is a MAC address? = [...] bits

VISUAL



6 bytes = 48 bits = 12 hexadecimal characters

ANSWER

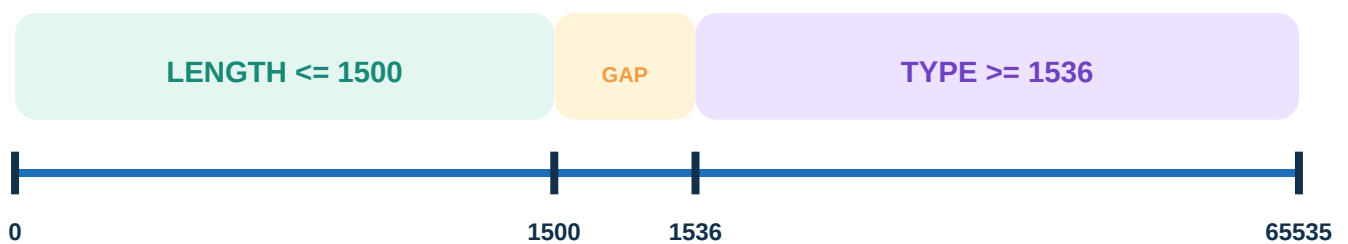
48

Memory cue: 48 bits = 6 bytes = 12 hex characters.

QUESTION

A value of [...] or less in the Type/Length field indicates Length.

VISUAL



ANSWER

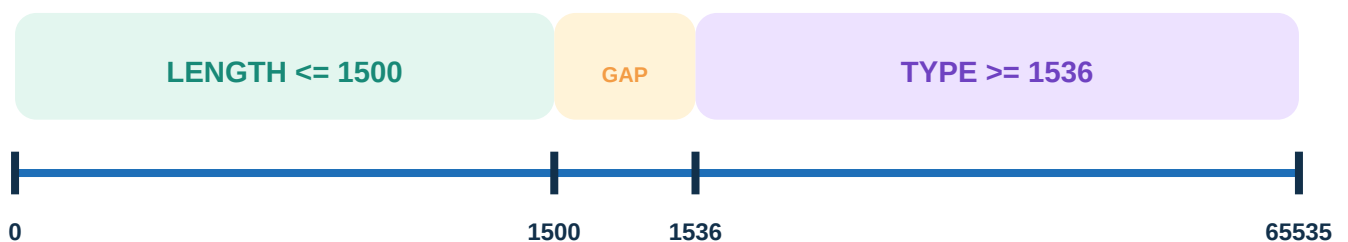
1500

Memory cue: 0-1500 means Length; 1501-1535 is intentionally unused for this distinction.

QUESTION

A value of [...] or greater in the Type/Length field indicates Type.

VISUAL



ANSWER

1536

Memory cue: 1536 decimal = 0x0600; values at/above this indicate EtherType.

QUESTION

What is the Ethernet Type value for IPv4? (in hexadecimal)

VISUAL

IPv4
0x0800

ANSWER

0x0800

Memory cue: EtherType IPv4: 0x0800.

QUESTION

What is the Ethernet Type value for IPv6? (in hexadecimal)

VISUAL

IPv6
0x86DD

ANSWER

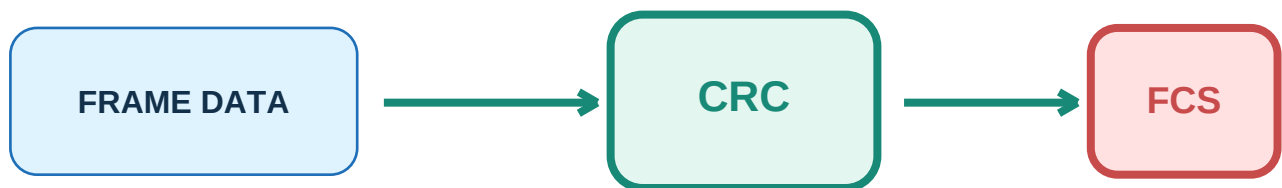
0x86DD

Memory cue: EtherType IPv6: 0x86DD.

QUESTION

What kind of algorithm does the Ethernet FCS use?

VISUAL



ANSWER

CRC (Cyclic Redundancy Check)

Memory cue: CRC calculates a check value; Ethernet stores it in the FCS field.

QUESTION

What is another name for a MAC address?

VISUAL



6 bytes = 48 bits = 12 hexadecimal characters

ANSWER

BIA (Burned-In Address)

Memory cue: BIA means the MAC address is burned into the network interface hardware.

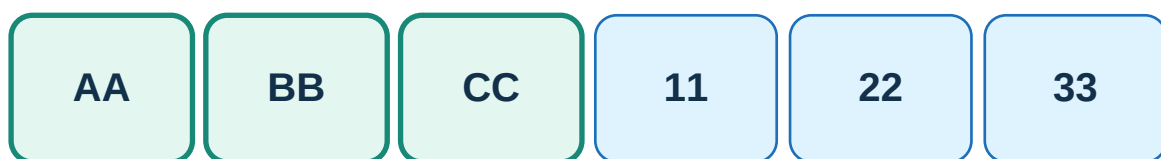
QUESTION

The first [...] bytes of a MAC address are the OUI.

VISUAL

OUI: first 3 bytes

Vendor/device-specific portion



6 bytes = 48 bits = 12 hexadecimal characters

ANSWER

3

Memory cue: OUI = first 24 bits / 3 bytes; it identifies the organization/vendor.

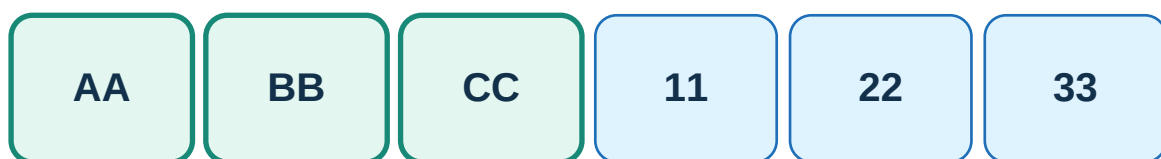
QUESTION

The first 3 bytes of a MAC address are the [...]

VISUAL

OUI: first 3 bytes

Vendor/device-specific portion



6 bytes = 48 bits = 12 hexadecimal characters

ANSWER

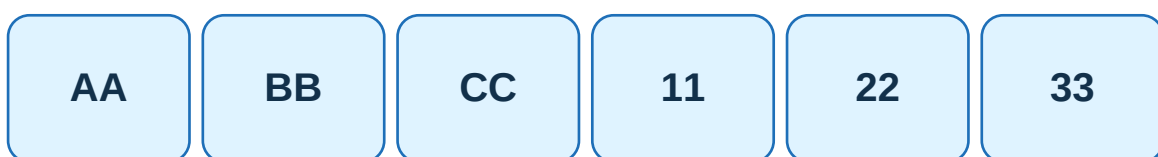
OUI (Organizationally Unique Identifier)

Memory cue: OUI = first 24 bits / 3 bytes; it identifies the organization/vendor.

QUESTION

A MAC address is written as [...] hexadecimal characters.

VISUAL



6 bytes = 48 bits = 12 hexadecimal characters

ANSWER

12

Memory cue: Each byte is two hex digits: $6 \text{ bytes} \times 2 = 12 \text{ hex characters}$.

QUESTION

Hexadecimal A = Decimal [...]

VISUAL

A 10	B 11	C 12	D 13	E 14	F 15
----------------	----------------	----------------	----------------	----------------	----------------

Hexadecimal digit -> decimal value

ANSWER

10

Memory cue: After 9, hexadecimal continues A=10 through F=15.

QUESTION

Hexadecimal B = Decimal [...]

VISUAL

A 10	B 11	C 12	D 13	E 14	F 15
----------------	----------------	----------------	----------------	----------------	----------------

Hexadecimal digit -> decimal value

ANSWER

11

Memory cue: After 9, hexadecimal continues A=10 through F=15.

QUESTION

Hexadecimal C = Decimal [...]

VISUAL

A	B	C	D	E	F
10	11	12	13	14	15

Hexadecimal digit -> decimal value

ANSWER

12

Memory cue: Each byte is two hex digits: 6 bytes $\times$ 2 = 12 hex characters.

QUESTION

Hexadecimal D = Decimal [...]

VISUAL

A	B	C	D	E	F
10	11	12	13	14	15

Hexadecimal digit -> decimal value

ANSWER

13

Memory cue: After 9, hexadecimal continues A=10 through F=15.

QUESTION

Hexadecimal E = Decimal [...]

VISUAL

A 10	B 11	C 12	D 13	E 14	F 15
----------------	----------------	----------------	----------------	----------------	----------------

Hexadecimal digit -> decimal value

ANSWER

14

Memory cue: After 9, hexadecimal continues A=10 through F=15.

QUESTION

Hexadecimal F = Decimal [...]

VISUAL

A	B	C	D	E	F
10	11	12	13	14	15

Hexadecimal digit -> decimal value

ANSWER

15

Memory cue: After 9, hexadecimal continues A=10 through F=15.

QUESTION

A [...] frame is a frame destined for a single target.

VISUAL

ETHERNET

Frame fields • MAC addressing • switching

ANSWER

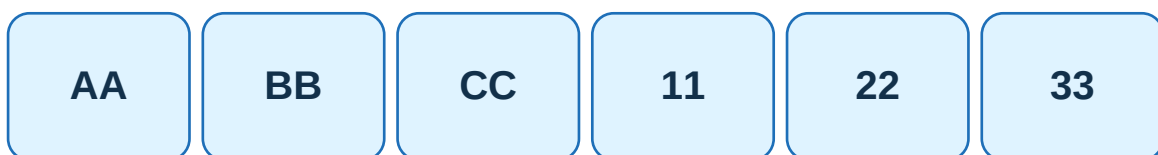
unicast

Memory cue: Connect this fact to the Ethernet frame and the switch MAC address table.

QUESTION

A MAC address learned by a switch without manual configuration is called a ...

VISUAL



6 bytes = 48 bits = 12 hexadecimal characters

ANSWER

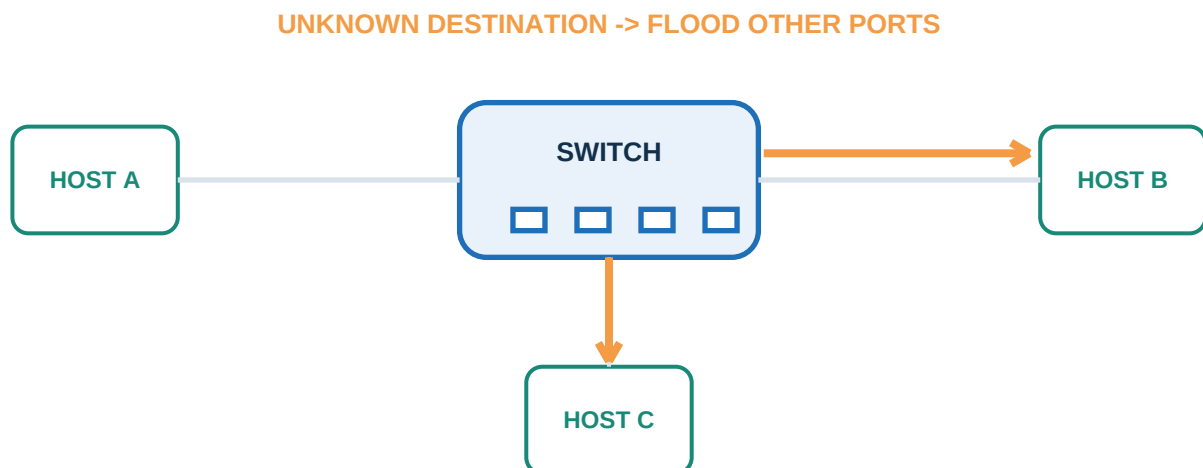
Dynamic MAC address (Dynamically learned MAC address)

Memory cue: A switch dynamically learns source MAC addresses from incoming frames.

QUESTION

What does a switch do with an unknown unicast frame?

VISUAL



ANSWER

Floods it out all ports (except the one it was received on)

Memory cue: Unknown destination? Flood to all other ports in the same broadcast domain/VLAN.

QUESTION

A switch will flood an [...] unicast frame.

VISUAL

ETHERNET

Frame fields • MAC addressing • switching

ANSWER

unknown

Memory cue: Known unicast is forwarded selectively; unknown unicast is flooded.

QUESTION

By default, a dynamic MAC address will be removed from the MAC address table after [time] of inactivity.

VISUAL



6 bytes = 48 bits = 12 hexadecimal characters

ANSWER

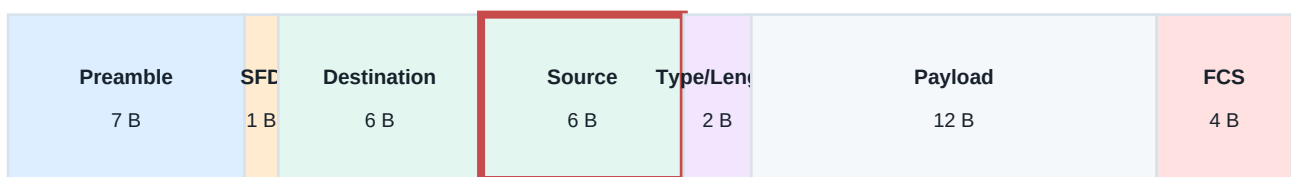
5 minutes

Memory cue: Default dynamic MAC aging timer in this deck: 5 minutes of inactivity.

QUESTION

Which field of an Ethernet frame does a switch use to learn and populate its MAC address table?

VISUAL



Simplified Ethernet frame layout (payload width compressed for study)

ANSWER

Source MAC Address

Memory cue: Learning is based on the SOURCE address; forwarding decisions use the DESTINATION address.

Day 05 - Quick Review

High-yield facts from the original 38-card deck

Ethernet frame

Header: Preamble 7 B | SFD 1 B | Destination 6 B | Source 6 B | Type/Length 2 B
Trailer: FCS 4 B. Preamble = synchronization; SFD = end of preamble.

Type / Length

≤ 1500 means Length. ≥ 1536 means Type. IPv4 EtherType = 0x0800. IPv6 = 0x86DD.

MAC address

6 bytes = 48 bits = 12 hex characters. Also called BIA. First 3 bytes = OUI.

FCS / CRC

FCS = Frame Check Sequence. Ethernet uses CRC (Cyclic Redundancy Check) for error detection.

Hexadecimal

A=10, B=11, C=12, D=13, E=14, F=15.

Switching

Switch learns SOURCE MAC on incoming port. Unknown unicast is flooded to all other ports.
Dynamic MAC entry ages out after 5 minutes of inactivity (per this deck).